

Multivariable Taylor polynomials

A calculation-first cheat sheet for approximating functions of several variables

The one-line idea. Near $a \in \mathbb{R}^n$, replace $f(a+h)$ by a polynomial in the displacement $h = x - a$. The gradient supplies the linear part; the Hessian supplies curvature.

$$T_2(a+h) = f(a) + \nabla f(a)^T h + \frac{1}{2} h^T H_f(a) h \quad \text{and} \quad f(a+h) = T_2(a+h) + \mathcal{O}(\|h\|^3).$$

01 / THE WORKFLOW

- Choose the centre:** write $a = (a_1, \dots, a_n)$ and $h = x - a$.
- Evaluate the function:** compute $f(a)$.
- Evaluate first derivatives:** compute $\nabla f(a) = (f_{x_1}(a), \dots, f_{x_n}(a))$.
- Evaluate second derivatives:** form the Hessian $H_f(a) = [f_{x_i x_j}(a)]$.
- Assemble:** use $f(a) + \nabla f(a)^T h + \frac{1}{2} h^T H_f(a) h$.
- State the remainder and domain:** degree 2 means error $\mathcal{O}(\|h\|^3)$; scalar prototypes may add conditions such as $|u| < 1$.
- Check one point:** compare T_2 with f numerically.

02 / FORMULAS TO MEMORISE

Object	Formula
First order	$T_1 = f(a) + \nabla f(a)^T h$
Second order	$T_2 = f(a) + \nabla f(a)^T h + \frac{1}{2} h^T H_f(a) h$
Two variables	$\frac{1}{2} h^T H h = \frac{1}{2} (f_{xx} u^2 + 2f_{xy} uv + f_{yy} v^2)_a$
Degree m	$T_m(a+h) = \sum_{ \alpha \leq m} \frac{D^\alpha f(a)}{\alpha!} h^\alpha$
Remainder	$f(a+h) - T_m(a+h) = \mathcal{O}(\ h\ ^{m+1})$

Multi-index key. $\alpha = (\alpha_1, \dots, \alpha_n)$, $|\alpha| = \sum_i \alpha_i$, $\alpha! = \prod_i \alpha_i!$, $D^\alpha f = \partial_1^{\alpha_1} \dots \partial_n^{\alpha_n} f$, and $h^\alpha = \prod_i h_i^{\alpha_i}$.

03 / WORKED EXAMPLE: EXPONENTIAL

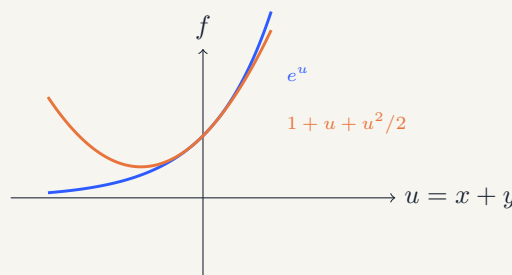
For $f(x, y) = e^{x+y}$ about $a = (0, 0)$, every first and second partial derivative equals e^{x+y} , so

$$f(a) = 1, \quad \nabla f(a) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad H_f(a) = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

With $h = (x, y)$,

$$T_2(x, y) = 1 + x + y + \frac{1}{2} (x^2 + 2xy + y^2) = 1 + x + y + \frac{1}{2} (x+y)^2.$$

At $(x, y) = (0.1, 0.2)$: $T_2 = 1.345$, whereas $e^{0.3} = 1.3498588\dots$; the absolute error is about 4.86×10^{-3} .



04 / BASIC WORKED EXAMPLES

Example 1: a quadratic is already its Taylor polynomial. Let $f(x, y) = x^2 + 3xy + y^2$ about $(0, 0)$. Since f has degree 2,

$$\nabla f(0, 0) = (0, 0), \quad H_f(0, 0) = \begin{pmatrix} 2 & 3 \\ 3 & 2 \end{pmatrix},$$

so $T_2(x, y) = x^2 + 3xy + y^2 = f(x, y)$. The remainder is exactly zero.

Example 2: a linear approximation. For $f(x, y) = \sqrt{x+y}$ about $(1, 1)$, write $u = x - 1$, $v = y - 1$. Since $f(a) = \sqrt{2}$ and $\nabla f(a) = (1/(2\sqrt{2}), 1/(2\sqrt{2}))$,

$$T_1(x, y) = \sqrt{2} + \frac{u+v}{2\sqrt{2}}.$$

At $(1.02, 0.97)$, $(u, v) = (0.02, -0.03)$ and $T_1 = 1.41068$; the exact value is $\sqrt{1.99} = 1.41067\dots$

Example 3: a vanishing quadratic term. For $f(x, y) = \sin(x+y)$ about $(0, 0)$, set $u = x + y$. Since $\sin u = u - u^3/6 + \dots$,

$$T_2(x, y) = x + y, \quad f(x, y) - T_2(x, y) = \mathcal{O}(\|(x, y)\|^3).$$

05 / INTERMEDIATE WORKED EXAMPLES

Example 4: a shifted logarithm. For $f(x, y) = \ln(x+y)$ about $(1, 1)$, let $u = x - 1$, $v = y - 1$. Then

$$\ln(x+y) = \ln(2+u+v) = \ln 2 + \frac{u+v}{2} - \frac{(u+v)^2}{8} + \mathcal{O}(\|(u, v)\|^3),$$

so $T_2 = \ln 2 + (u+v)/2 - (u+v)^2/8$. The condition is $|u+v| < 2$.

Example 5: reciprocal with cross terms. For $f(x, y) = 1/(1-x-y)$ about $(0, 0)$, use $(1-s)^{-1} = 1+s+s^2+s^3+\dots$ with $s = x+y$:

$$T_3 = 1 + (x+y) + (x+y)^2 + (x+y)^3.$$

Expanded, the cubic part is $x^3 + 3x^2y + 3xy^2 + y^3$. The convergence condition is $|x+y| < 1$.

06 / WORKED EXAMPLE: A NONZERO CENTRE

Let $f(x, y) = x^2e^y$ and expand about $a = (1, 0)$. Use displacement variables $u = x - 1$ and $v = y$.

$$f(1, 0) = 1, \quad \nabla f(1, 0) = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad H_f(1, 0) = \begin{pmatrix} 2 & 2 \\ 2 & 1 \end{pmatrix}.$$

The quadratic form is

$$\frac{1}{2} \begin{pmatrix} u & v \end{pmatrix} \begin{pmatrix} 2 & 2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = u^2 + 2uv + \frac{1}{2}v^2.$$

Therefore

$$T_2(x, y) = 1 + 2u + v + u^2 + 2uv + \frac{1}{2}v^2, \quad u = x - 1, \quad v = y.$$

At $(x, y) = (1.1, 0.05)$, $(u, v) = (0.1, 0.05)$ and $T_2 = 1.25625$; the exact value $1.1^2e^{0.05} = 1.256889\dots$, so the error is about 6.39×10^{-4} .

07 / ADVANCED WORKED EXAMPLES

Example 6: degree four from a composite monomial. For $f(x, y) = e^{xy}$ about $(0, 0)$, set $s = xy$. Since s has total degree 2,

$$e^{xy} = 1 + xy + \frac{(xy)^2}{2} + \mathcal{O}(\|(x, y)\|^6), \quad T_4(x, y) = 1 + xy + \frac{1}{2}x^2y^2.$$

The next term $(xy)^3/6$ has total degree 6, so there are no degree-5 terms.

Example 7: radial dependence. For $f(x, y) = \ln(1+x^2+y^2)$ about $(0, 0)$, let $r^2 = x^2 + y^2$:

$$T_4(x, y) = r^2 - \frac{1}{2}r^4 = x^2 + y^2 - \frac{1}{2}(x^2 + y^2)^2,$$

with validity condition $x^2 + y^2 < 1$.

08 / FAST PATTERNS

Scalar-composition shortcut. If $f(x, y) = \phi(u)$ with $u = ax + by$ and ϕ has a one-variable series, expand in u first, then substitute $u = ax + by$.

$$\ln(1+u) = u - \frac{u^2}{2} + \mathcal{O}(u^3), \quad e^u = 1 + u + \frac{u^2}{2} + \mathcal{O}(u^3), \quad (1+u)^\alpha = 1 + \alpha u + \frac{\alpha(\alpha-1)}{2}u^2 + \mathcal{O}(u^3).$$

09 / WORKED EXAMPLE: LOGARITHM

Let $f(x, y) = \ln(1 + x + 2y)$ near $(0, 0)$. Set $u = x + 2y$ and use $\ln(1+u) = u - u^2/2 + \mathcal{O}(u^3)$:

$$T_2(x, y) = x + 2y - \frac{1}{2}(x + 2y)^2 = x + 2y - \frac{1}{2}x^2 - 2xy - 2y^2.$$

The validity condition is $|x + 2y| < 1$. For $(0.1, -0.05)$, $u = 0$, so both the exact function and T_2 equal 0.

10 / WORKED EXAMPLE: SQUARE ROOT

For $g(x, y) = \sqrt{1 + x - y}$, set $u = x - y$ and use $(1+u)^{1/2} = 1 + u/2 - u^2/8 + \mathcal{O}(u^3)$:

$$T_2(x, y) = 1 + \frac{1}{2}(x - y) - \frac{1}{8}(x - y)^2.$$

At $(0.2, 0.1)$, $u = 0.1$ and $T_2 = 1.04875$, while $\sqrt{1.1} = 1.0488088\dots$

11 / ERROR AND VALIDITY

Situation	What to write
Local degree-2 approximation	$f = T_2 + \mathcal{O}(\ h\ ^3)$
Known third-derivative bound M	$ R_2 \leq M\ h\ ^3/3!$ (with a compatible norm and neighbourhood)
Alternating one-variable tail	The first omitted term bounds the tail when term magnitudes decrease
Composition $\phi(ax + by)$	Inherit the scalar convergence condition on $u = ax + by$

Cross term reminder. If $f_{xy}(a) \neq 0$, the quadratic term contains $f_{xy}(a)uv$ after the factor $\frac{1}{2}$ is distributed. Omitting the cross term changes the surface tilt along diagonal directions.

12 / COMMON PITFALLS

- Using x and y instead of displacements $u = x - a_1$ and $v = y - a_2$.
- Forgetting that the Hessian is evaluated at the centre.
- Writing $f_{xy}uv/2$ instead of $f_{xy}uv$ in the two-variable quadratic form.
- Giving a polynomial without its remainder order or convergence condition.
- Confusing a Taylor polynomial (finite) with the full Taylor series (an infinite limit).

13 / FINAL CHECKLIST

1. Centre a stated? ✓
2. $f(a)$, gradient, Hessian evaluated? ✓
3. Displacement $h = x - a$ substituted? ✓
4. Degree and remainder stated? ✓
5. Domain or scalar-series condition stated? ✓
6. One numerical check performed? ✓